The Chinese University of Hong Kong, Shenzhen

Ordinary Differential Equation Notes

Author: Shuo Guo

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1 First Order Differential Equations

For the first order differential equations, we would consider to solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

Consider a simpler case:

$$\frac{\mathrm{d}}{\mathrm{d}} = g(t)$$

One with elementary calculus knowledge can solve it:

$$y(t) = \int g(t)dt + c$$

However, for general first order equation, it's very hard to solve.
We would first consider some simpler cases.

1.1 First order linear differential equations

Consider f is consisted by linear equations.

Definition 1.1 (First-oder linear differential equation)**.**

The general first-order linear differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{1}$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. To slove general first order linear equation(1), we could consider a simpler case:

Definition 1.2 (Homogeneous first order linear equation)**.**
The equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0 \tag{2}$$

is called homogeneous first-order linear differential equation

Now we try to solve them part by part.

1.1.1 Homogeneous first order linear differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0.$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)y.$$
$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t).$$

Integrating both sides with respect to t gives

$$\int \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}\,dt = -\int a(t)\,dt.$$

By the chain rule, $\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$, hence

$$\ln|y(t)| = -\int a(t)\,dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^{C} e^{-\int a(t)\,dt}.$$

Since $e^{C} > 0$, we may absorb the sign into a new constant $c = \pm e^{C}$, obtaining the general solution

$$\boxed{y(t) = c\,e^{-\int a(t)\,dt}} \tag{3}$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

In most cases, we are not interest in all solutions of (3). We could add more assmuption as

$$y(t_0) = y_0 \tag{4}$$

Then we integral (3) between t_0 and t.
Recalling $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$, we could write:

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}s}\ln|y(s)|ds = \int_{t_0}^{t} -a(s)ds$$

So

$$\ln|y(t)| - \ln|y(t_0)| = \ln|\frac{y(t)}{y(t_0)}| = -\int_{t_0}^{t} a(s)ds$$

Thus,

$$|\frac{y(t)}{y(t_0}| = \exp(-\int_{t_0}^{t} a(s)ds)$$

The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.
So

$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds) \tag{5}$$

1.1.2 General first order linear differential equation

Consider the problem:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{6}$$

We want to write it in a form:

$$\frac{\mathrm{d}'something'}{\mathrm{d}t} = c(t)$$

and then we can solve it. Multiply both sides by a continuous function $\mu(t)$ [1]
So

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)\mu(t)y = \mu(t)b(t)$$

We consider another formula: $\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \frac{\mathrm{d}y}{\mathrm{d}t}\mu(t) + y\frac{\mathrm{d}\mu(t)}{\mathrm{d}t}$.
So we try to choose proper μ s.t. the left of the equation can be written as $\frac{\mathrm{d}'something'}{\mathrm{d}t}$.
We choose

$$\mu(t) = \exp(\int a(t)dt) \tag{7}$$

And

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\mu(t)y = \int \mu(t)b(t)dt + c$$

$$y = \frac{1}{\mu(t)}(\int \mu(t)b(t)dt + c), \quad \text{with } \mu(t) = \exp(\int a(t)dt) \tag{8}$$

We could consider $y(t_0) = y_0$ So:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\Downarrow$$

$$\mu(t)y - \mu(t_0)y_0 = \int_{t_0}^{t} \mu(s)b(s)ds$$

Thus,

$$y = \frac{1}{\mu(t)}\Big(\mu(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)dt\Big) \tag{9}$$

[1]here μ could somehow be without more limit. So we could choose what we want later to make up form we want

1.2 Separable equations

Recall how we deal with linear differential equation:

$$\frac{1}{y(t)}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$$

$$\Downarrow$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln|g(t)| = -a(t)$$

So for a more general ODE with the form:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{g(t)}{f(y)} \tag{10}$$

We could write it as

$$\frac{\mathrm{d}}{\mathrm{d}t}F(y(t)) = g(t), \quad \text{with } F(y) = \int f(y)dy$$

So the answer is:

$$F(y(t)) = \int g(t)dt + c \tag{11}$$

We could also add an assumption $y(t_0) = y_0$
So

$$F(y(t)) - F(y(t_0)) = \int_{t_0}^{t} g(s)ds$$

$$\int_{y_0}^{y} f(r)dr = \int_{t_0}^{t} g(s)ds$$

1.3 Population models

Let $p(t)$ denotes the population of a given species at time t, $r(t,p)$ denotes the difference between its birth rate and death rate.
For the simplest, we think that r will not change with t and p, which means it's a constant. The simplest model:

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = ap(t), \quad a \text{ is a constant}$$

Also, assume $p(t_0) = p_0$
We can easily find the solution

$$p(t) = p_0 \exp(a(t - t_0)) \tag{12}$$

This result works very well if the population is small. However, if the population grows larger and larger, it falls. So we can let

$$\frac{\mathrm{d}r}{\mathrm{d}t} = ar - br^2 \tag{13}$$

It's also a sparable problem, though the integer part is a bit complex (but absolutly solvable!)
The answer is:

$$p(t) = \frac{ap_0 e^{a(t-t_0)}}{a - bp_0 + bp_0 e^{a(t-t_0)}} = \frac{ap_0}{bp_0 + (a - bp_0)e^{-a(t-t_0)}}.$$

1.4 Exact equations and the range of equations we can solve

Theorem 1.1. *Let $M(t, y)$ and $N(t, y)$ be continuous and have continuous partial derivatives with respect to t and y in the rectangle R consisting of those points (t, y) with $a < t < b$ and $c < y < d$. There exists a function $\phi(t, y)$ such that*

$$M(t, y) = \frac{\partial \phi}{\partial t} \quad \textit{and} \quad N(t, y) = \frac{\partial \phi}{\partial y}$$

if, and only if,

$$\frac{\partial M}{\partial y} \Big/ \frac{\partial N}{\partial t} = 1$$

in R.